KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5333C | 5 Credits :4 | Program core | Source-grounded

IoT and Applications

Understand the role of IoT in various domains of Industry.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5333C is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5333C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automation and Robotics
Course code	5333C
Course title	IoT and Applications
Semester	5 Credits :4
Course category	Program core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

8 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Understand the role of IoT in various domains of Industry.

Course outcomes

CO	Official outcome	Study level
CO1	14 networking. Perceive ideas on the different connectivity Understanding	See official syllabus
CO2	14 technologies and Wireless Sensor Networks. Explain the basic concepts of cloud computing Understanding	See official syllabus
CO3	15 and fog computing Apply the basics of IoT for different applications Understanding	See official syllabus
CO4	15 like Smart Homes and Industrial IoT.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1
CO2	CO2 2 2
CO3	CO3 2 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the basic concepts of IoT and IoT networking.	2	As specified
Module 2	Sensor Networks.	2	As specified
Module 3	Explain the basic concepts of cloud computing and fog computing	2	As specified
Module 4	Industrial IoT.	2	As specified

MODULE 1

Explain the basic concepts of IoT and IoT networking.

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basic concepts of IoT and IoT networking.
- Official duration: As specified
- Official topic entries captured: 2

- The full source excerpt is preserved below for coverage checking.

Define IoT 6 Understanding Enlist any three characteristics of IoT and

Define IoT 6 Understanding Enlist any three characteristics of IoT and is an official topic in Module 1 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define IoT 6 Understanding Enlist any three characteristics of IoT and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define iot 6 understanding enlist any three characteristics of iot and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define IoT 6 Understanding Enlist any three characteristics of IoT and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Define IoT 6 Understanding Enlist any three characteristics of IoT and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless

8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless is an official topic in Module 1 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding explain. characteristics of iot- applications of iot- iot categories- iot enablers and connectivity layers - baseline technologies-sensors-characteristics of a sensor classification of sensors-actuators-types of actuators-iot components and implementation. service oriented architecture-iot interdependencies-challenges for iot. iot networking connectivity terminologies, gateway prefix allotment-impact of mobility on addressing multihoming-deviations from regular web-iot identifications and data protocols. perceive ideas on the different connectivity technologies and wireless should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the basic concepts of IoT and IoT networking.

    M1.01     Define IoT                                          6
Understanding
                Enlist any three characteristics of IoT and
    M1.02                                                          8
Understanding
                explain.
  Contents: Introduction to Internet of Things Introduction
  Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and
  Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor
  Classification of Sensors-Actuators-Types of Actuators-IoT components and
  implementation. Service Oriented Architecture-IoT Interdependencies-Challenges
for IoT.
  IoT Networking
  Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on
Addressing
  Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols.
                Perceive ideas on the different connectivity technologies and
Wireless
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Sensor Networks.

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Sensor Networks.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

8 Understanding HART.

8 Understanding HART. is an official topic in Module 2 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding HART. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding hart. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding HART. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 8 Understanding HART. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN.

Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. is an official topic in Module 2 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the challenges in wsn 6 understanding introduction-ieee 802.15.4-zigbee-6lowpan-rfid-hart and wireless hart-nfc bluetooth-z-wave-isa 100.11a. wireless sensor networks introduction-components of a sensor node-modes of detection-challenges in wsn- sensor web-cooperation-behaviour of nodes in wsn-information theoretic self- management of wsn-social sensing in wsn-applications of wsn-wireless multimedia sensor networks. wireless nano sensor networks-under water acoustic sensor networks-wsn coverage. optimal geographical density control (ogdc) algorithm- stationary wsn-mobile wsn. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN-Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary

WSN-Mobile WSN. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Sensor Networks.
                Discuss on Plasma HART and Wireless
    M2.01                                                         8
Understanding
                HART.
    M2.02     Explain the challenges in WSN                      6
Understanding

              Series Test – I                                    1
  Contents: Connectivity Technologies
  Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC
  Bluetooth-Z-wave-ISA 100.11A.
  Wireless Sensor Networks
  Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN-
  Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self-
  Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia
  Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor
  Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm-
  Stationary WSN-Mobile WSN.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the basic concepts of cloud computing and fog computing

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basic concepts of cloud computing and fog computing
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Explain public cloud. 8 nding Understa

Explain public cloud. 8 nding Understa is an official topic in Module 3 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain public cloud. 8 nding Understa using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain public cloud. 8 nding understa should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain public cloud. 8 nding Understa means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain public cloud. 8 nding Understa □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and

List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and is an official topic in Module 3 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list out the advantages of fog. 7 nding introduction-architecture-characteristics-deployment models-public cloud-private cloudhybrid cloud-community cloud-multi cloud-distributed cloud-inter cloud-big data cloud-hpc cloud-service models-service management-cloud security. fog computing introduction-why fog computing-requirements of iot-architecture of fog-working of fog-advantages of fog-applications of fog-challenges in fog. apply the basics of iot for different applications like smart homes and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the basic concepts of cloud computing and fog computing


Understa
   M3.01      Explain public cloud.                                    8

nding

Understa
   M3.02      List out the advantages of Fog.                          7

nding
  Contents: Cloud Computing
  Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private
  CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big
  Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security.
  Fog Computing
  Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of
  Fog-Advantages of Fog-Applications of Fog-Challenges in Fog.
              Apply the basics of IoT for different applications like Smart Homes
and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Industrial IoT.

This module is presented from the official syllabus scope for IoT and Applications. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Industrial IoT.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Understand non-industrial applications of IoT 7

Understand non-industrial applications of IoT 7 is an official topic in Module 4 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand non-industrial applications of IoT 7 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand non-industrial applications of iot 7 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand non-industrial applications of IoT 7 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Understand non-industrial applications of IoT 7 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT-Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications

Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT-Manufacturing Industry-Health Care Service Industry-Transportation and Logistics-Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications is an official topic in Module 4 of IoT and Applications. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT-Manufacturing Industry-Health Care Service Industry-Transportation and Logistics-Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand industrial applications of iot 8 smart homes introduction-origin of smart home-examples of smart home technologies-smart home implementation-home area networks-han elements-han standards-han architectures -han initiatives-smart home benefits and issues. industrial iot introduction-iiot requirements-design considerations-applications of iiot- manufacturing industry-health care service industry-transportation and logistics- mining-firefighting-smart dust-drones-futuristic farming-aerospace-energy networks- benefits of iiot-challenges of iiot. text / reference t/r book title/author honbo zhou, “the internet of things in the cloud: a middleware perspective”, t1 crc press,2012 t2 jeeva jose, “internet of things”, khanna book publishing co.(p) ltd, 1/e hanes david ,salgueiro gonzalo , grossetete patrick , barton rob , henry jerome, r1 “iot fundamentals: networking technologies, protocols and use cases for the internet of things”, pearson, 1/e. rajkamal, “internet of things: architecture and design principles”, mcgraw r2 hill (india) private limited daniel minoli,

— “building the internet of things with ipv6 and mipv6: the r3 evolving world of m2m communications”, isbn: 978-1-118-47347-4, willy publications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT-Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India)

Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Industrial IoT.

   M4.01      Understand non-industrial applications of IoT                  7

   M4.02      Understand industrial applications of IoT                      8

               Series Test – II                                       1
  Contents: IoT Applications
  Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home
  Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN
  Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues.
  Industrial IoT
  Introduction-IIoT     Requirements-Design   Considerations-Applications  of
IIoT-
  Manufacturing Industry-Health Care Service Industry-Transportation and
Logistics-
  Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy
Networks-
  Benefits of IIoT-Challenges of IIoT.

Text / Reference

   T/R                                    Book Title/Author
          Honbo Zhou, “The Internet of Things in the Cloud: A Middleware
Perspective”,
    T1
          CRC Press,2012
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define IoT 6 Understanding Enlist any three characteristics of IoT and. (3 marks)

Answer: Begin with the standard definition or identity of *Define IoT 6 Understanding Enlist any three characteristics of IoT and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain 8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding explain. Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols. Perceive ideas on the different connectivity technologies and Wireless*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□□ □□□□□□□.

Module 2

Define or explain 8 Understanding HART.. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding HART.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the challenges in WSN 6 Understanding Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN- Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self- Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm- Stationary WSN-Mobile WSN.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Explain public cloud. 8 nding Understa. (3 marks)

Answer: Begin with the standard definition or identity of *Explain public cloud. 8 nding Understa*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and. (3 marks)

Answer: Begin with the standard definition or identity of *List out the advantages of Fog. 7 nding Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private CloudHybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog. Apply the basics of IoT for different applications like Smart Homes and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Understand non-industrial applications of IoT 7. (3 marks)

Answer: Begin with the standard definition or identity of *Understand non-industrial applications of IoT 7*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications. (3 marks)

Answer: Begin with the standard definition or identity of *Understand industrial applications of IoT 8 Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT- Manufacturing Industry-Health Care Service Industry-Transportation and Logistics- Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks- Benefits of IIoT-Challenges of IIoT. Text / Reference T/R Book Title/Author Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, T1 CRC Press,2012 T2 Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, R1 “IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things”, Pearson, 1/e. Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw R2 Hill (India) Private Limited Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The R3 Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define IoT 6 Understanding Enlist any three characteristics of IoT and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **8 Understanding HART.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain public cloud. 8 nding Understa**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Understand non-industrial applications of IoT 7**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson